Tutorial 5

1. A binary random variable X takes on only two values with probabilities

$$P[X = 0] = p \qquad \text{and} \qquad P[X = 1] = 1 - p.$$

Find $E[X]$, the variance of X, and $E[X^5]$.

2. For a non-negative random variable X, we know that

$$E[X] = 3 \qquad \text{and} \qquad \text{Var}(X) = 1.$$

(a) Find

$$E\left[(3X - 2)^2\right].$$

(b) Find

$$\text{Var}(2X + 1).$$

3. For a continuous random variable, prove that the mean defined by

$$E[X] = \int_{-\infty}^{\infty} x f_X(x)\, dx$$

can also be calculated directly using the CDF of X using the following formula:

$$E[X] = \int_0^{\infty} (1 - F_X(t))\ dt - \int_{-\infty}^{0} F_X(t)\, dt.$$

Hint: You can use

$$\frac{d}{dt} F_X(t) = f_X(t), \qquad \int u\, dv = uv - \int v\, du,$$

and assume that

$$\lim_{t\to\infty} t\,(1 - F_X(t)) = 0 \qquad \text{and} \qquad \lim_{t\to-\infty} t F_X(t) = 0.$$

4. For a normally distributed random variable X with mean 12 and variance 4:

(a) Find c such that $P[X \geq c] = 0.5$.

(b) Find $E[X^2]$.

5. A random variable X has the following pdf

$$f_X(x) = \begin{cases} Cx^2, & \text{for } |x| \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find C, and cdf of X.

(b) Determine the mean and variance of X.

(c) Find the pdf of $Y = |X|$, $E[Y^2]$, and $\text{Var}[3X - 1]$.